

# **Geometrically Inductive Self-Supervised Learning for EEG**

A Design Study of Spatial Attention Across Sessions, Subjects, and Montages

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# Problem: EEG is non-stationary – and the geometry is what changes

EEG varies on **three geometric axes**:

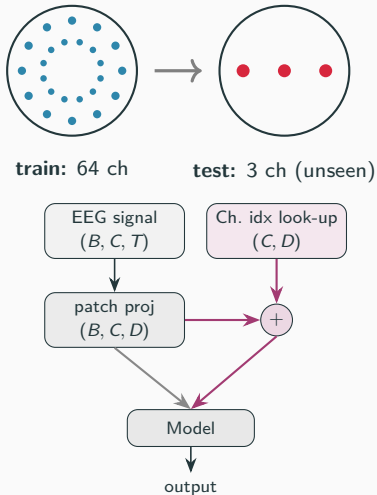
- **Session** — cap shifts day to day.  
(*motivates*)
- **Subject** — head sizes differ.  
(*motivates*)
- **Montage** — 2 to 256 electrodes in different layouts. ← **what I test**

**Codex models** (LaBraM, EEGPT):

identity-indexed embeddings — **no way to generalize** to an unseen layout.

**Channel-independent** (BIOT): no spatial structure at all.

Neither uses scalp geometry as an *input*.



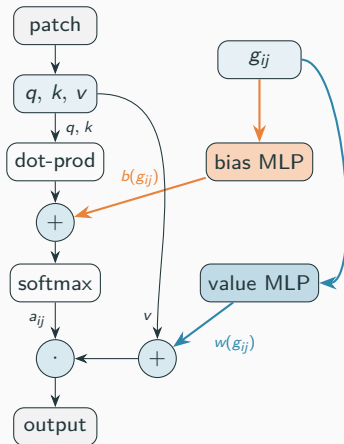
## Solution: spatial attention conditioned on $g_{ij}$

$$g_{ij} = [p_i, p_j, p_i - p_j, \|p_i - p_j\|] \in \mathbb{R}^{10}$$

**Example:**  $g_{C3 \rightarrow Cz}$  (normalized coords)

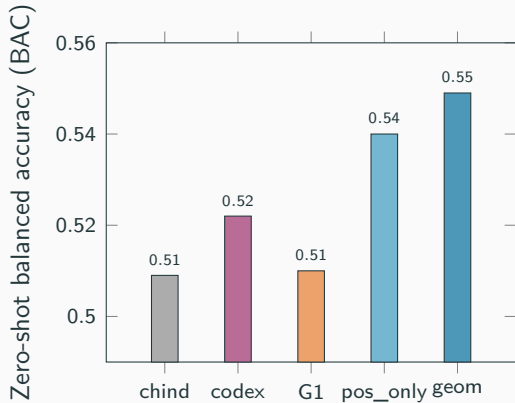
|                 |             |                        |
|-----------------|-------------|------------------------|
| $p_i$ (C3)      | $(x, y, z)$ | $(-0.71, 0.00, 0.59)$  |
| $p_j$ (Cz)      | $(x, y, z)$ | $(0.00, 0.00, 0.95)$   |
| $p_i - p_j$     | disp.       | $(-0.71, 0.00, -0.36)$ |
| $\ p_i - p_j\ $ | dist.       | 0.80                   |

$3 + 3 + 3 + 1 = 10$  dims. **pos\_only** ablation  
keeps just  $[p_i, p_j] \in \mathbb{R}^6$ .



**bias path** only = G1, **value path** only = G2, both = G3.

# Headline: at scale, geometric beats both baselines on zero-shot transfer



78 PhysioNet + 78 Sleep-EDFx pretrain → zero-shot to BCIC-2B

cells = row-normalized recall (%);  
diag mean = BAC

**geom** (G2)

|     | Pred L | Pred R |
|-----|--------|--------|
| T.L | 56.4   | 43.6   |
| T.R | 46.6   | 53.4   |

**pos\_only** (G3)

|     | Pred L | Pred R |
|-----|--------|--------|
| T.L | 53.8   | 46.2   |
| T.R | 45.8   | 54.2   |

# Conclusion: a clean positive, with clear edges

## What I found:

- Geometry-conditioned attention **transfers zero-shot** to an unseen montage and **beats both baselines**.
- Robust to descriptor reduction (pos\_only also significant).
- The signal lives in the **value path**; the **bias path** (G1) alone barely moves accuracy — less significant, but can still help.
- The **codex baseline**, even given name-aligned transfer and a regime that favors it, stays near chance — it cannot generalize to an unseen montage.

## Honest limits:

- Demonstrated on **one** held-out montage (BCIC); broader montage coverage remains open.

**Future work:** second distinct-montage held-out (e.g. 128 ch); larger corpus.

## Did I run what the proposal promised?

| Proposed experiment                   | Outcome                                       |
|---------------------------------------|-----------------------------------------------|
| E2(c) — cross-montage zero-shot       | ✓ <b>the headline win, significant</b>        |
| E3 — where $g_{ij}$ enters (G1/G2/G3) | ✓ <b>G2 &gt; G3 <math>\gg</math> G1</b>       |
| E4 — channel-independent baseline     | ✓ <b>promoted to my baseline</b>              |
| E1 — learned-codex baseline           | ✓ <b>name-aligned; near chance</b>            |
| E2(a) — cross-session                 | session drift is physiological, not geometric |
| E2(b) — cross-subject / head size     | no public head-size-varied dataset            |

**Top:** delivered. **Bottom:** interim conclusion for the feasibility study.

# Key References

- Wang et al. **EEGPT**. NeurIPS 2024 – EEG foundation model; training recipe.
- Jiang et al. **LaBraM**. ICLR 2024 – codex (identity-indexed) baseline.
- Yang et al. **BIOT**. NeurIPS 2023 – channel-independent baseline.
- Schalk et al. 2004 – PhysioNet MI dataset.
- Tangermann et al. 2012 – BCIC-2B dataset.
- Kemp et al. 2000 – Sleep-EDFx dataset.

Full reference list and per-experiment details in the project report.